Lecture 30.03: Gradient Boosting Regressor — Implementation & Tuning











📌 Topic Overview

Practical implementation of GradientBoostingRegressor in scikit-learn. Explores synthetic regression data generation, key hyperparameters including learning_rate, n_estimators, subsample, and max_depth, along with R2 evaluation.

📋 Key Learning Objectives

- Generating synthetic regression datasets using make_regression.
- Importing and configuring GradientBoostingRegressor from sklearn.ensemble.
- Crucial Hyperparameters: learning_rate, n_estimators, subsample, and max_depth.
- Stochastic Gradient Boosting: Using subsample < 1.0 to reduce tree variance.
- Evaluating regression quality with R2 score and Mean Squared Error.

🔍 In-Depth Conceptual Breakdown

- **Synthetic Data Generation**: `make_regression(n_samples=1000, n_features=10, n_informative=5, random_state=42)` se realistic regression data banaya gaya.
- **Train-Test Split**: `train_test_split(X, y, test_size=0.2, random_state=42)` se 80% training aur 20% testing data alag kiya gaya.
- **Hyperparameter 1:** `learning_rate` : Shrinkage factor (default = 0.1). Chota learning rate (0.01 - 0.1) model ko overfit hone se bachata hai.
- **Hyperparameter 2:** `n_estimators` : Kitne sequential trees banane hain (default = 100). Chote learning rate ke sath zyada trees lagte hain.

- **Hyperparameter 3:** `subsample` : **Stochastic Gradient Boosting** enable karta hai. Jaise `subsample=0.8` ka matlab har tree random 80% samples par train hoga.
- **Hyperparameter 4:** `max_depth` : Har weak tree ki depth (default = 3). Trees ko shallow rakha jata hai taaki individual trees overfit na karein.
- **Model Evaluation**: Test set par R^2 score aur `mean_squared_error` calculate karke generalization performance verify ki gayi.

📊 Architectural Flowchart



🔢 Mathematical Formulation

The mean squared error objective function:

$$\text{MSE} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2$$

The coefficient of determination R^2:

$$R^2 = 1 - \frac{\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{N}(y_i - \bar{y})^2}$$

💻 Implementation

The fully tested and verified standalone implementation is provided in the synchronized Jupyter Notebook.